Business Intelligence & Data Analytics | Internship Project | June 2026

INTRODUCTION

Retail businesses operate in high-competition, margin-sensitive environments where data-driven decisions are critical for sustained profitability. This project was undertaken to analyze transactional retail data and extract actionable business intelligence across product categories, seasons, and inventory behavior. The end goal was to equip stakeholders with a clear, visual understanding of what drives and drains retail profitability.

ABSTRACT

A retail dataset of **2,000 transaction records** spanning 5 product categories, 4 regions, and 4 seasons was analyzed using SQL, Python, and Tableau. Data was cleaned in Python (Pandas) to handle 187 missing values via median imputation. SQL aggregations revealed category and seasonal profitability patterns. A Pearson correlation (r = –0.007) between inventory days and profit confirmed no linear relationship between holding duration and margins. Key findings: **Furniture** led margins at 25.23%; **Winter** generated the highest revenue (■44.6M); **Snacks and Dairy** ranked as the top sub-categories. Findings were presented via an interactive Tableau dashboard with Region, Season, and Category filters.

■133M	■31M	22.58%	150.53K	-0.007
Total Revenue	Total Profit	Avg Margin	Units Sold	Inv. Corr. (r)

TOOLS USED

Tool	Purpose
SQL — MySQL Workbench	Database creation, aggregations, profitability & seasonal queries
Python — Pandas, NumPy	Data loading, type validation, null imputation, correlation analysis
Tableau Desktop	Interactive KPI dashboard with Region, Season & Category filters
CSV Dataset	Primary data source — 2,000 retail transaction records (14 fields)

STEPS INVOLVED IN BUILDING THE PROJECT

Step 1	**Data Collection & Inspection** Loaded retail_data_raw.csv (2,000 records, 14 fields) using Pandas. Inspected data types, shape, and summary statistics. Identified 187 missing values across inventory_days, discount_pct, and profit_margin_pct columns. Confirmed zero duplicate records.
Step 2	**Data Cleaning (Python)** Imputed missing inventory_days and discount_pct values using column medians — chosen over zero-fill to preserve statistical integrity. Recalculated missing profit_margin_pct values from existing profit and revenue fields. Added inventory_status (Fast/Slow Moving) and profit_status (Profit/Loss) classification columns. Exported cleaned data as retail_data_clean.csv.
Step 3	**SQL Database Setup & Querying** Imported cleaned CSV into MySQL Workbench (retail_analysis database). Executed queries for: category-wise profit margin analysis, sub-category profit ranking, seasonal revenue and margin comparison, and inventory days vs profit aggregation by sub-category.
Step 4	**Python Correlation Analysis** Computed Pearson correlation between inventory_days and profit: r = –0.007. Concluded that inventory holding duration has no meaningful linear relationship with profitability, recommending product-level analysis over turnover-duration heuristics.
Step 5	**Tableau Dashboard Development** Built an interactive dashboard with KPI tiles (Revenue ■133M, Profit ■31M, Margin 22.58%, Units 150.53K), Profit by Season bar chart, Profit by Category and Sub-Category horizontal bars, and an Inventory Days vs Profitability scatter plot. Added Region, Season, and Category filters for stakeholder exploration.

CONCLUSION

This project demonstrated how a structured, multi-tool BI pipeline can transform raw retail data into actionable business intelligence. **Furniture (25.23%)** and **Sports (24.44%)** emerged as the highest-margin categories, while **Clothing** requires pricing strategy review. **Winter** is the peak revenue season, demanding proactive inventory planning. The correlation analysis (r = –0.007) disproved the assumption that slow-moving stock is necessarily unprofitable — a key insight for inventory management. Together, the SQL analysis, Python statistics, and Tableau dashboard equip retail decision-makers with a clear, evidence-based framework for improving profitability, optimizing inventory allocation, and capitalizing on seasonal demand cycles.